

Extraction of features from signals on cell-graphs

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Intro (purpose of the project):

- Starting from a dataset of various patients and their status

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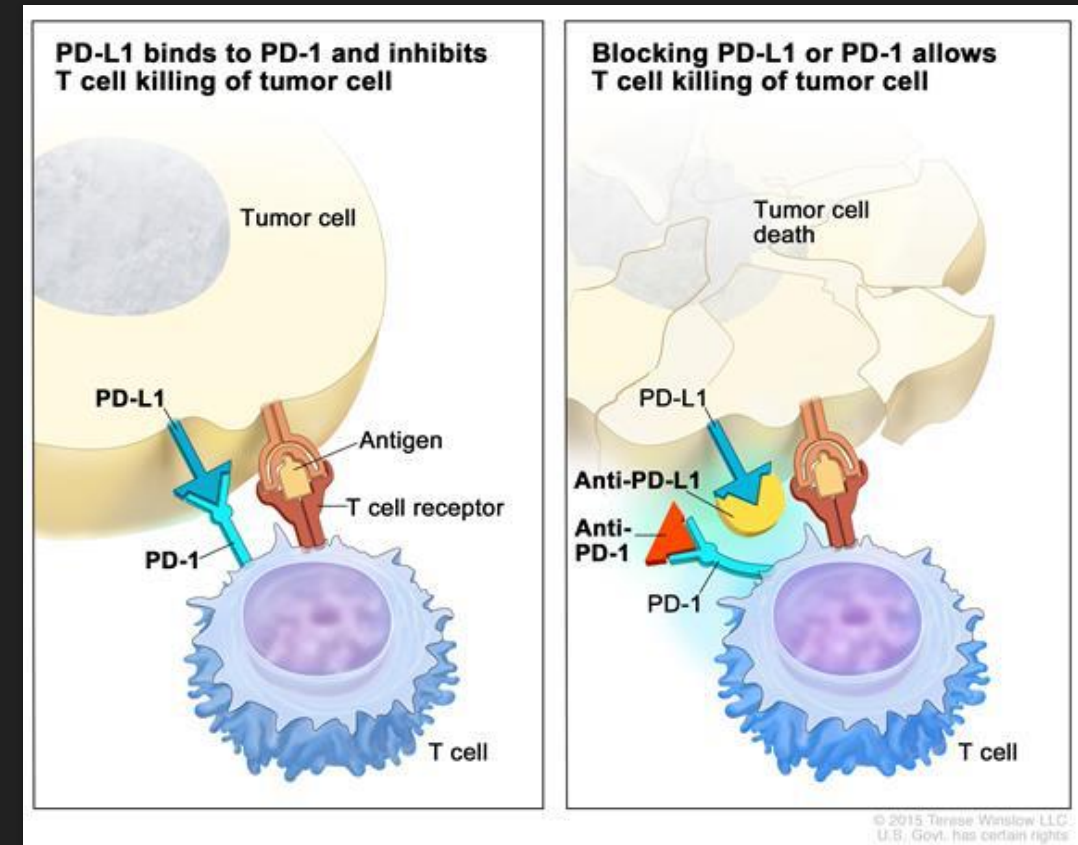
- Starting from a dataset of various patients and their status
- Find elements that make a patient more reactive than another
- Interpret a tissue of cells as a graph in order to extract features
- Features that are still interpretable (Eg.: “Diameter” or “Degree distribution”)

Intro (Environment):

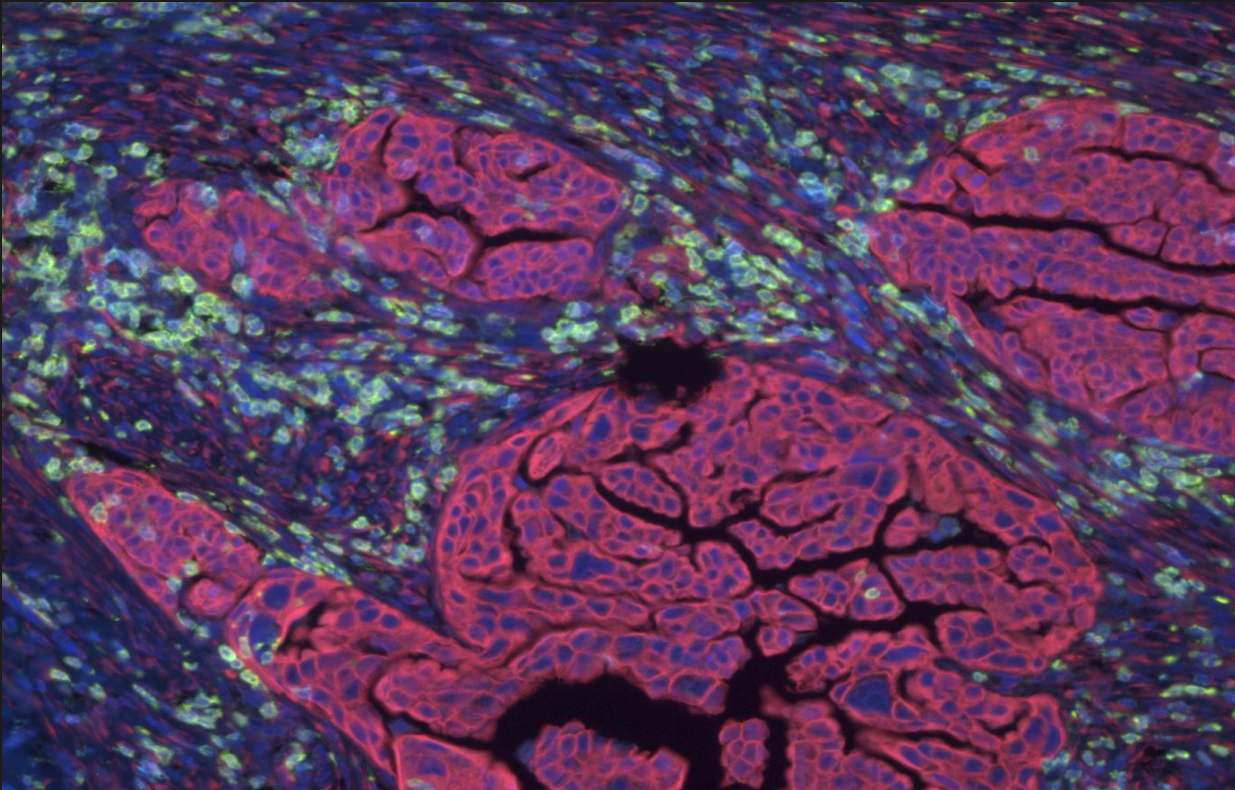
- The environment from which the dataset for this project was extracted is called Tumor micro environment (TME) or Stroma micro environment and it is composed of cells (fibroblasts, endothelial cells, inflammatory cells like the macrophages, neutrophils, lymphocytes) + extra-cellular substance (proteins), inside is located the tumor vascularization
- Various Biomarkers used to identify the cells and their type. We used those biomarkers as signal over the graph, in particular:
 - Biomarkers: any substance, structure, or process that can be measured in the body or its products and influence or predict the incidence of outcome or disease

Intro (Environment):

Here is possible to see (as an example of signal and why are them helpful) how the PD-1/PD-L1 system interact in order, for T-cells, to recognize (or not) an hazard to attack

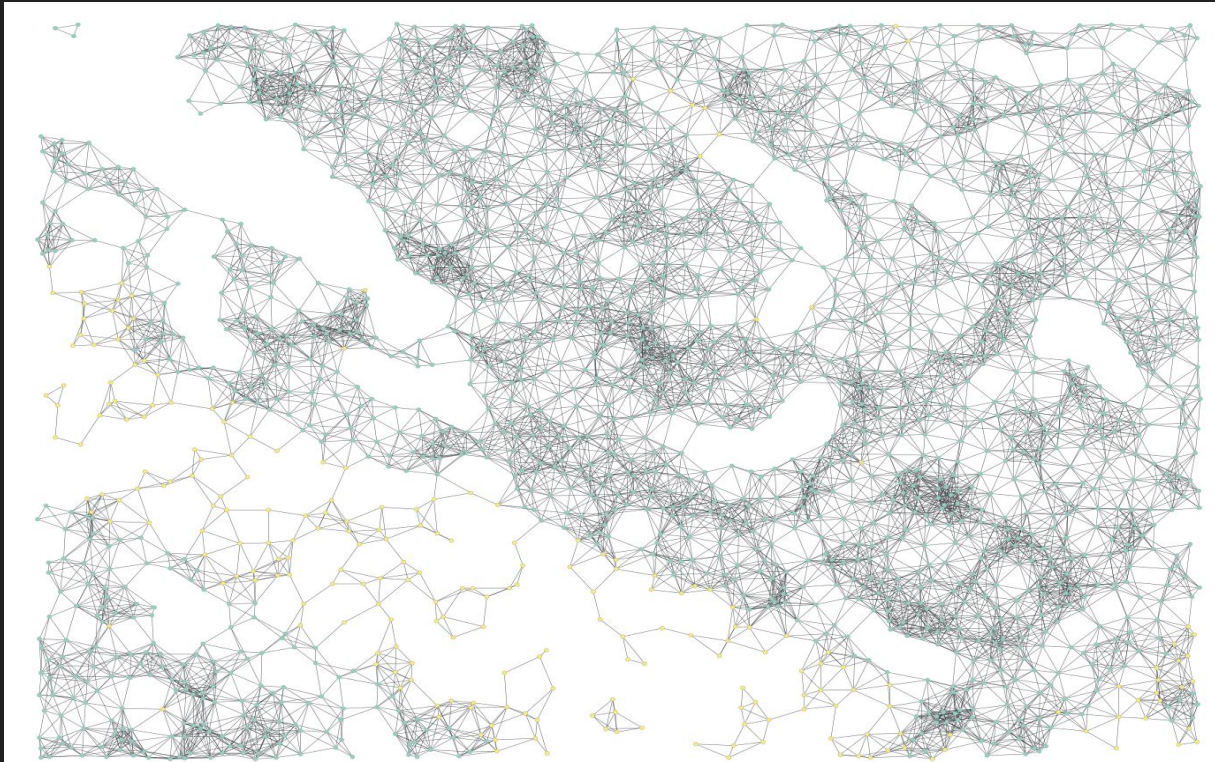


Tasks:



- Input: A tissue of cells
- Parsed to obtain graphs:
 - Full graph
 - Stroma graph
 - Tumor graph
- 4 Signals defined on the graph:
 - Area
 - Keratin
 - PD1
 - Category
- Features computed on this graph
- Correlation between these features can provide information about the patient

Types of features:



5 Types of features

- Connectedness: clustering, isolated point, etc..
- Distances: radius, diameter, Closeness, etc..
- Spectral: algebraic multiplicity of certain eigenvalues (0, 1, 2), trace, etc..
- Signal over graph: smoothness, spectrum, etc..
- PD1

List of features:

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'isolatedNode': 0.001694915254237288},
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'avgDiameter90': 1253.5808523003654,
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'largestEignW': 513.8015824101507,
'secondLargestEignW': 505.58147672776045,
'traceL': 1180.0}}}
```

58 features in total

- Organized in a dictionary object
- Not always of trivial definition

PD1:

PD1 related feature:

- Penetration of the PD1:
mean distance of PD1-positive cells to the next stroma (hence, how much is penetrated in the tumour)

```
{'full': {'PD1': {'PD1penetration': 0},
```

```
{'full': {'PD1': {'PD1penetration': 0},
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```

Connectedness:

connectedness related feature:

- Various features related to how nodes are interconnected, eg.:
 - Average degree
 - Giant Component ratio
 - Etc...
- Remain very interpretable

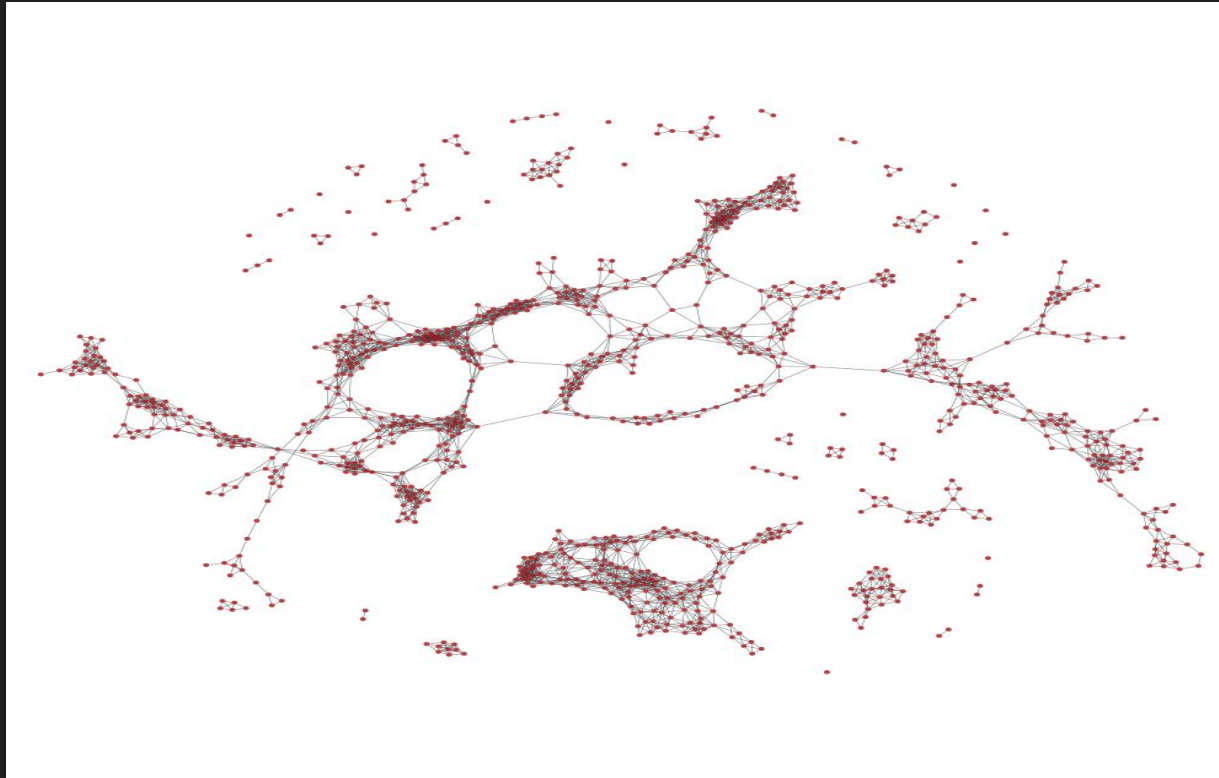
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Problem with distances:

The graphs obtained are composed of multiple, disconnected components

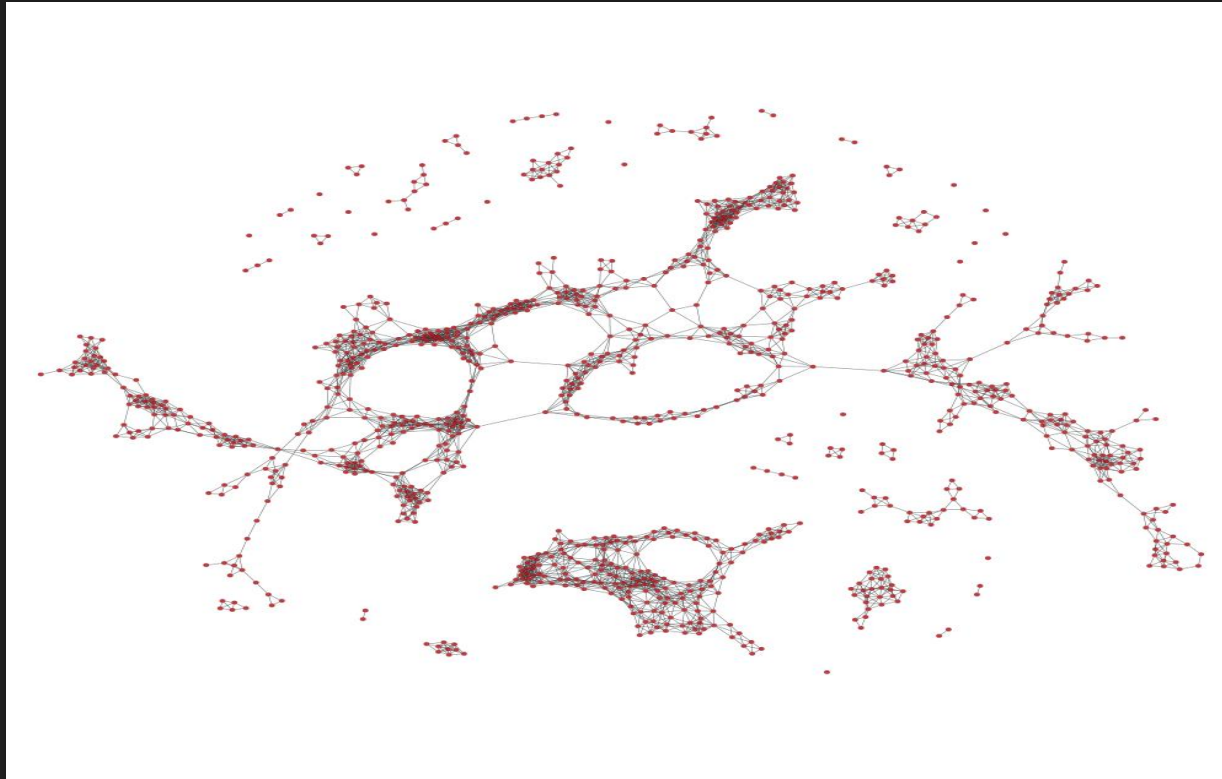


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            'traceL': 1180.0}}},
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```

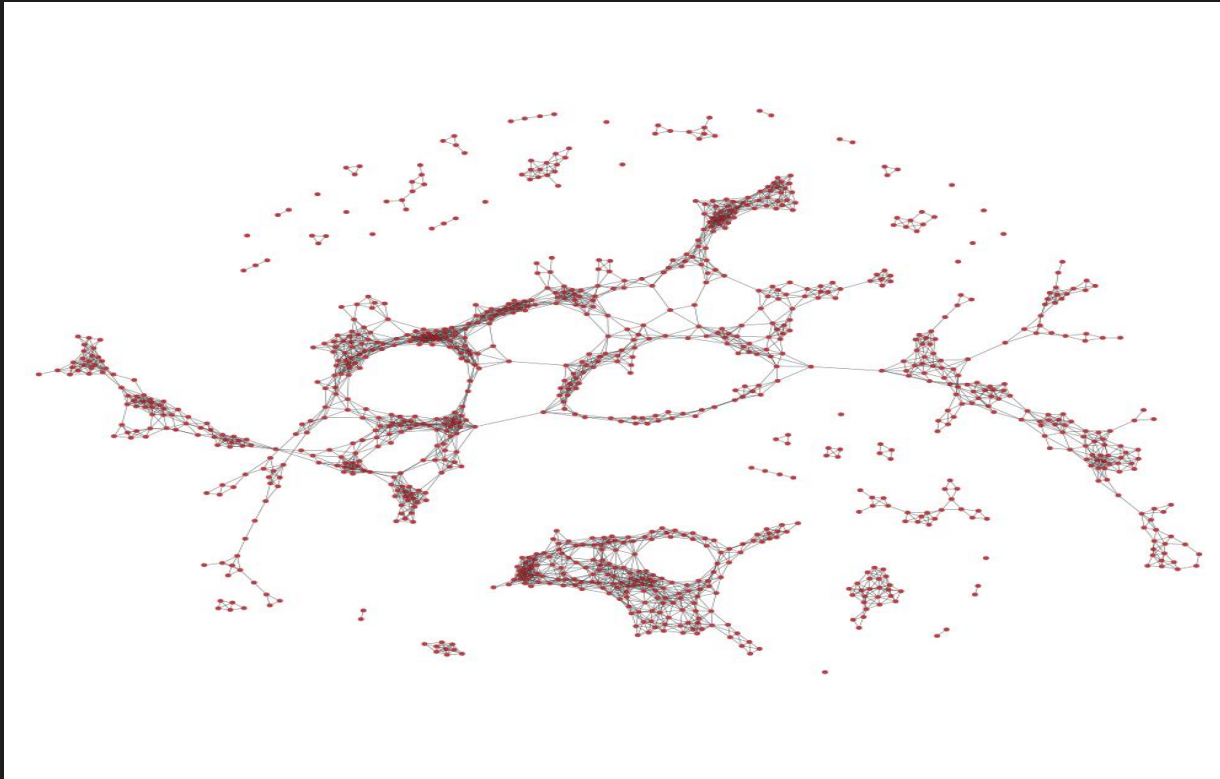
Problem with distances:



The graphs obtained are composed of multiple, disconnected components

The distance in a graph is defined for connected graphs and by definition is, otherwise, infinity

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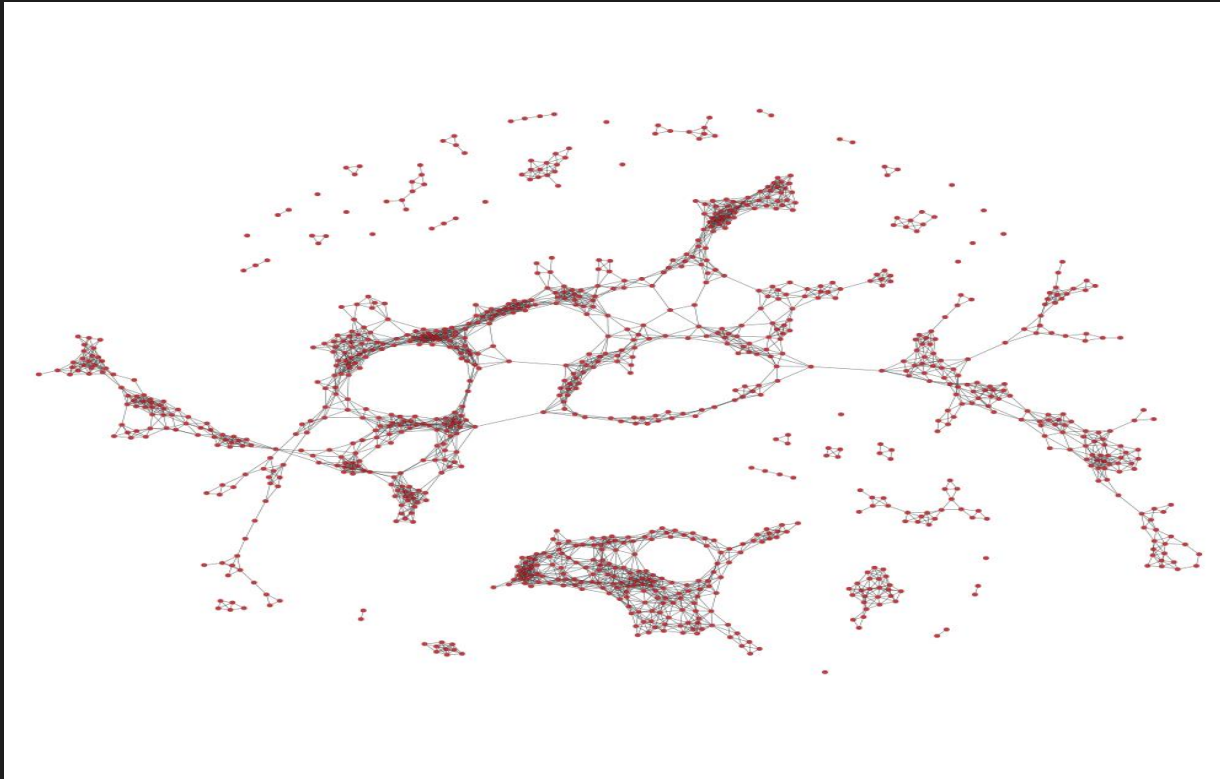


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IDEA: divide the graph in subgraph of meaningful dimension and compute all the distances metrics for them

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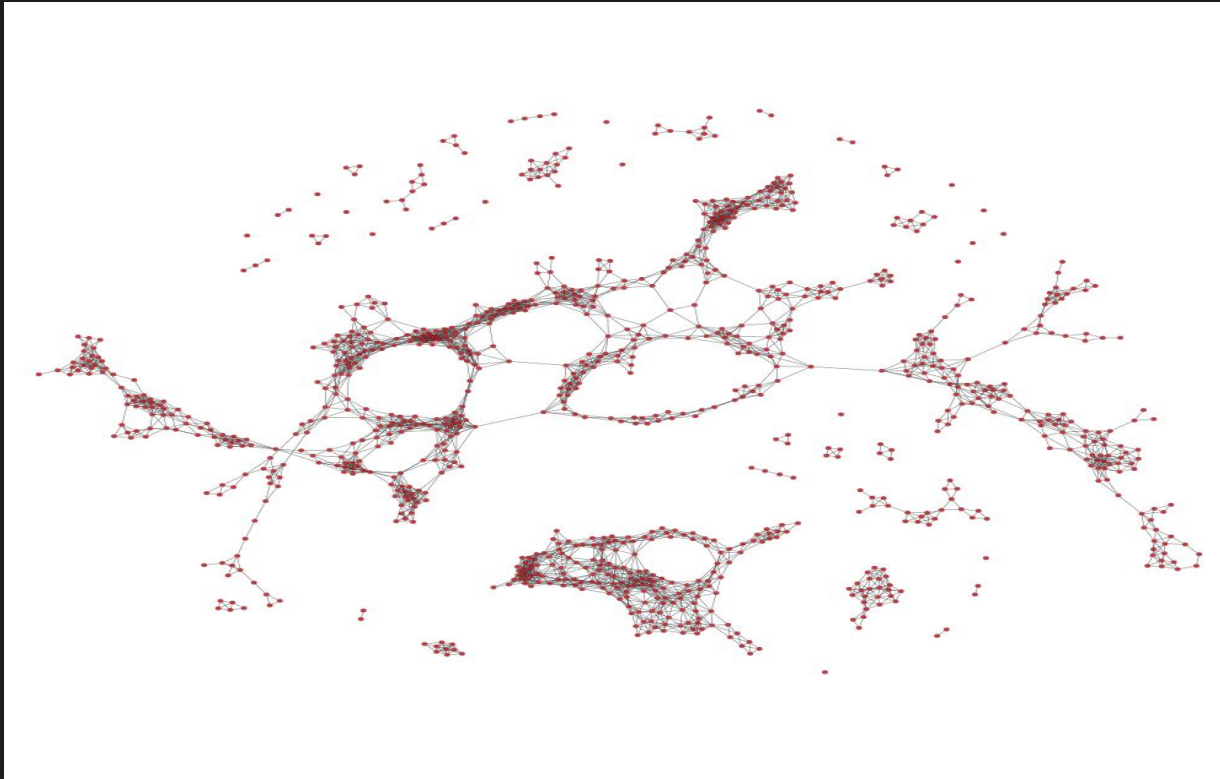
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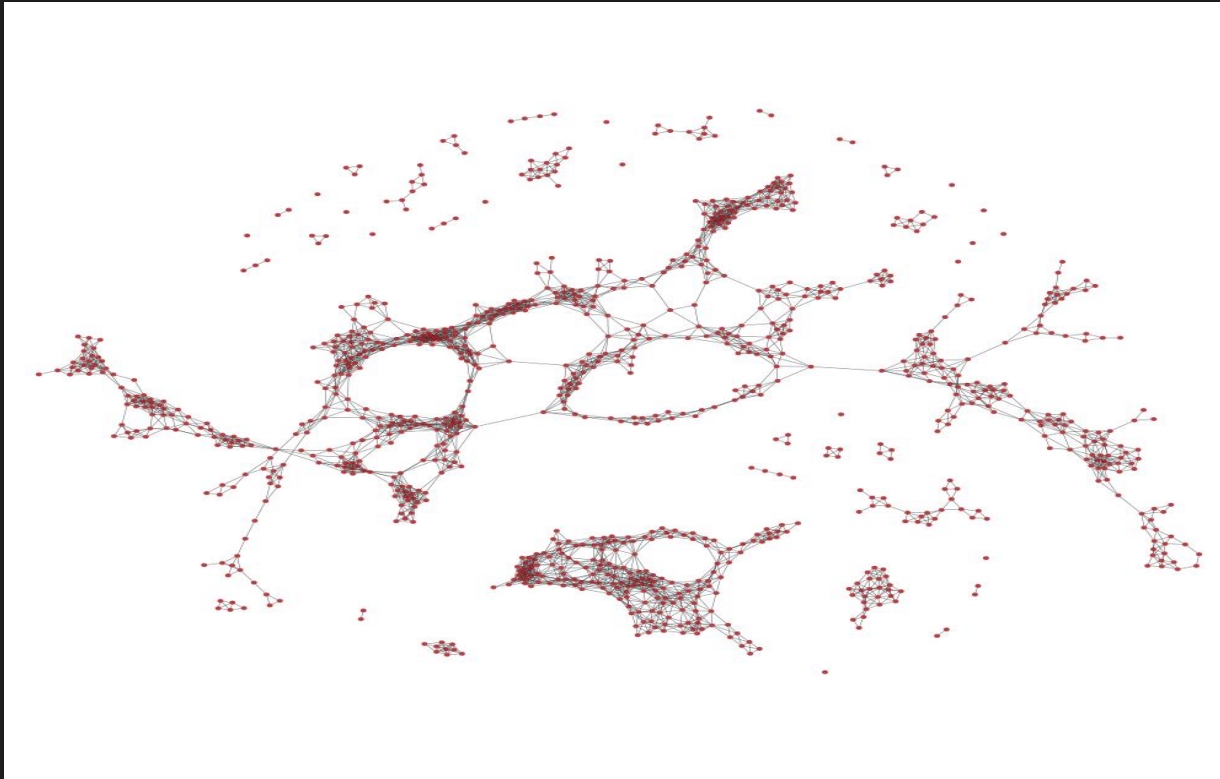
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A simple average over the obtained results would be nonsense

Weight every features of a subgraph with the number of nodes in it:

Problem with distances:



The Feature F_i then is:

$$F_i = \sum_{k=0}^{N-1} f_{ik} * r_k$$

With:

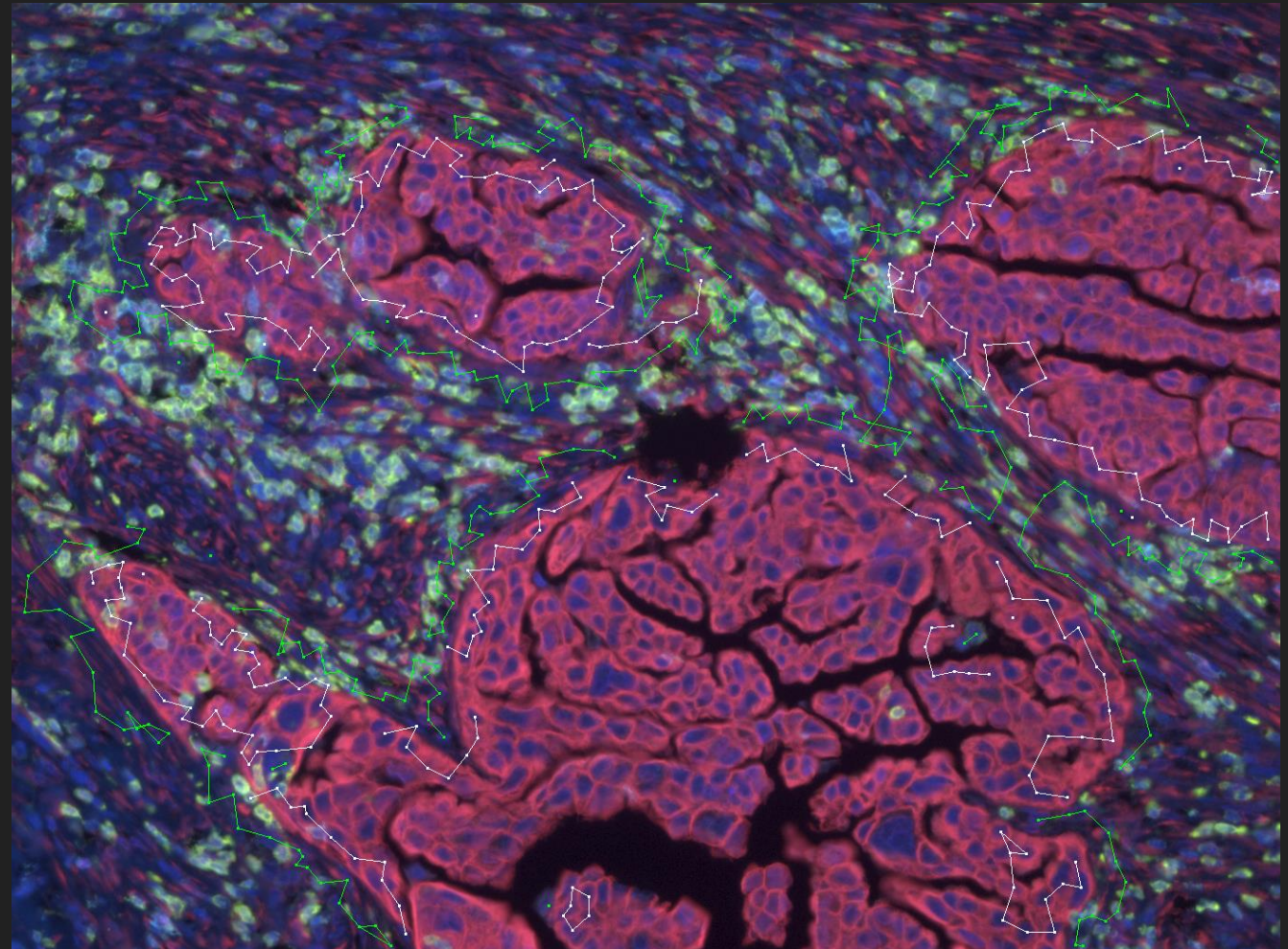
f_{ik} : Feature i for component k

r_k : $\frac{\text{Node in component } k}{\text{Total number of node}}$

Length of the cut:

Special mention to the length of the cut (distance category):

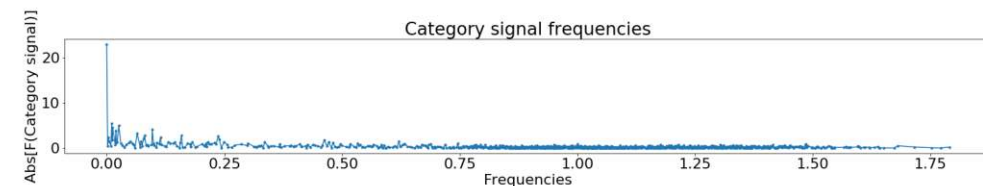
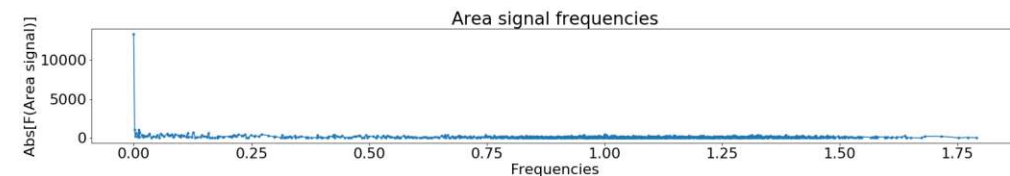
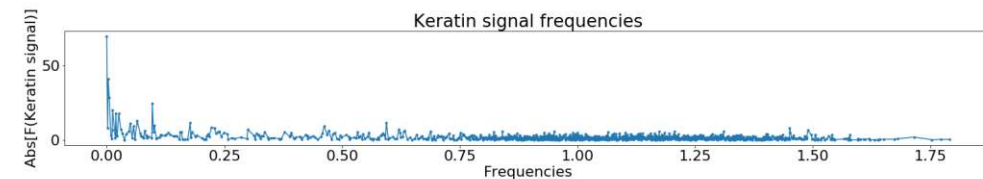
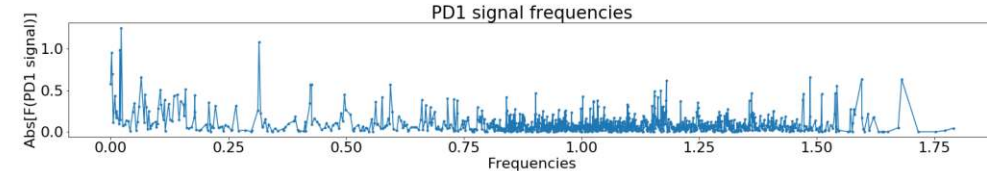
- At first compute chunks of border directly on the edge (in contact with cells of the opposite type)
- Then connect the borders with the one nearer to them (within a range) recursively



Signal:

Features Based on the frequency decomposition of the signal on the graph:

- First 3 value (low frequencies) of the Graph F. Transform for each signal
- First 3 value (low frequencies) of the filtered Graph F. Transform for each signal
- Is the set-up of a potential workbench:
 - We can see that the PD1 signal has high values also in the high freq.
 - Optimal will be to consider the whole spectrum as input to machine-learning algorithm



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'spectral': {'eigen0L': 1,
'eigen1L': 2,
'eigen2L': 0,
'energyL': 1392400.0,
'largestEigenW': 513.8015824101507,
'secondLargestEigenW': 505.58147672776945,
'traceL': 1180.0}}}
```

Spectral:

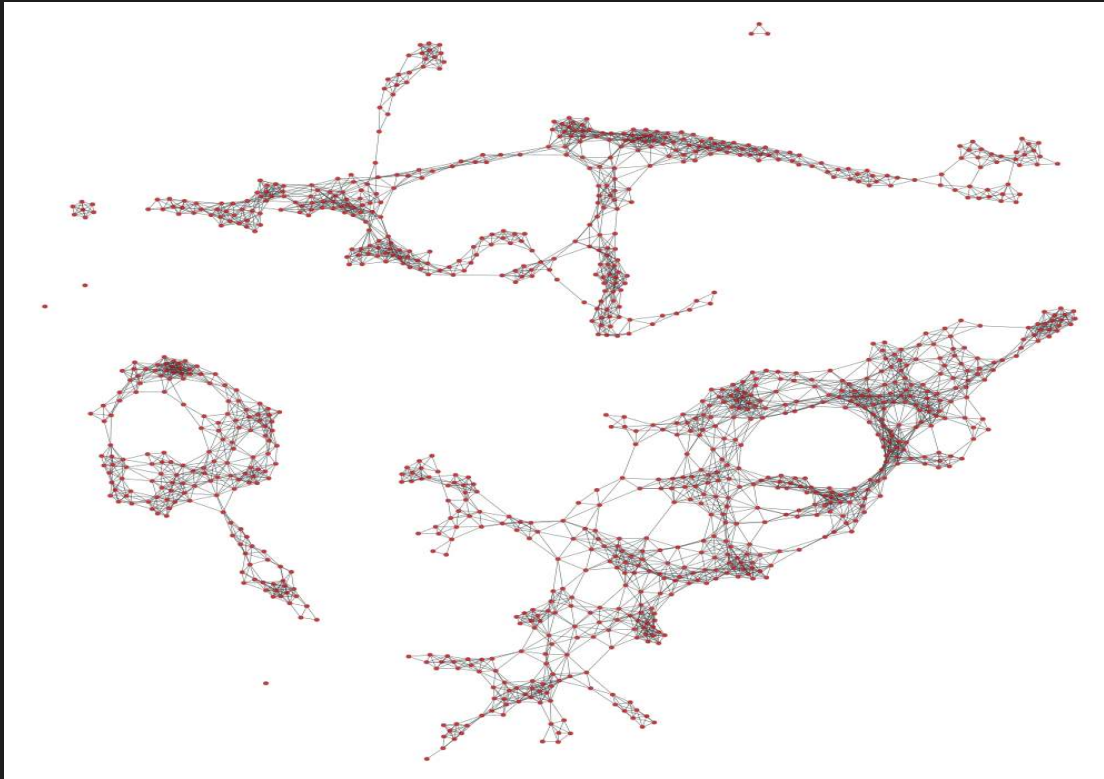
Features related to the eigenvalues on the matrix representation of the graph:

- Certain values of the eigenvalues of the Laplacian matrix
- Lose in interpretability

```
'spectral': {'eigen0L': 1,
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'energyL': 1392400.0,
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'keratinSecondFreqFilt': 0.31700313249871104,
'keratinThirdFreq': 41.16871445141019,
'keratinThirdFreqFilt': 0.32911093622726045},
'spectral': {'eigen0L': 1,
'eigen1L': 2,
'eigen2L': 0,
'energyL': 1392400.0,
'largestEignW': 513.8015824101507,
'secondLargestEignW': 505.58147672776045,
'traceL': 1180.0}}}
```

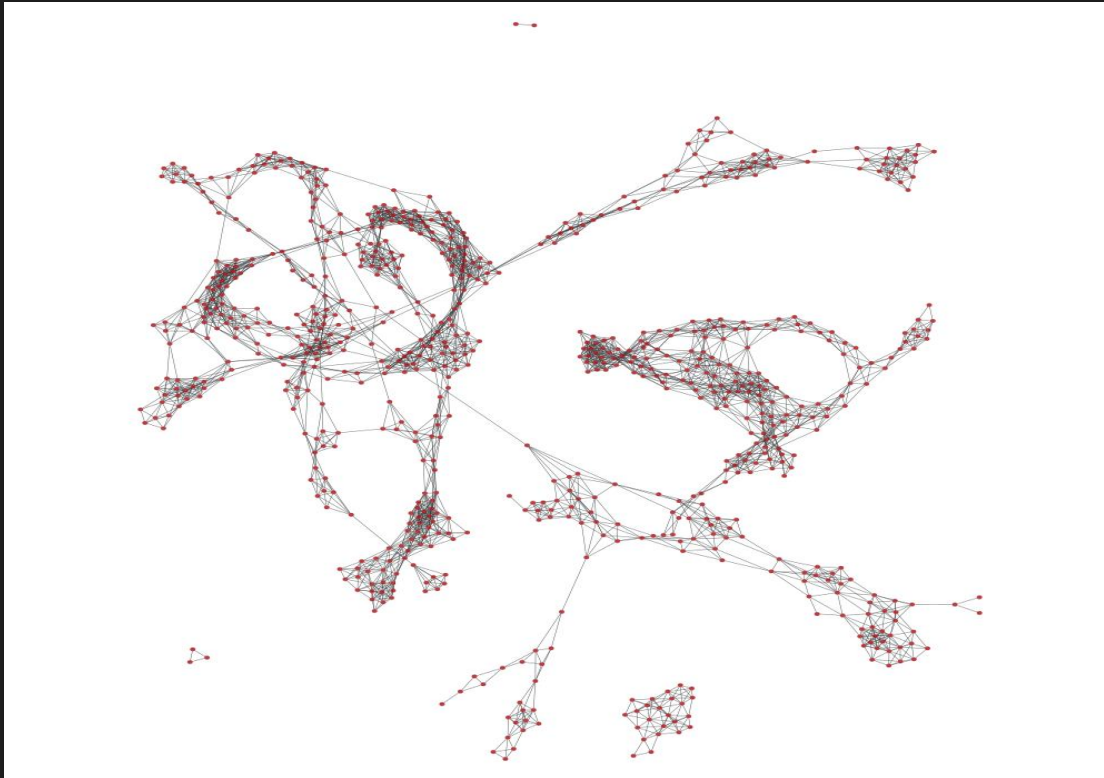
Some features:



Graph: Tumor 3:

- Ratio of the giant component :
$$R_{gc} = \frac{\text{Nodes in the biggest component}}{\text{Total number of nodes}} = 0.485$$
- Average closeness :
$$Cl_{avg} = \frac{1}{\sum_{i=1}^N \sum_{j=1, j \neq i}^N d_{ij}} = 381.57$$
- Average eccentricity (at 90%) :
$$Ec_{avg} = \frac{1}{N} \sum_{i=1}^N \max(d_{ij}, \forall j \in [1, N] - \{i\}) = 846.20$$

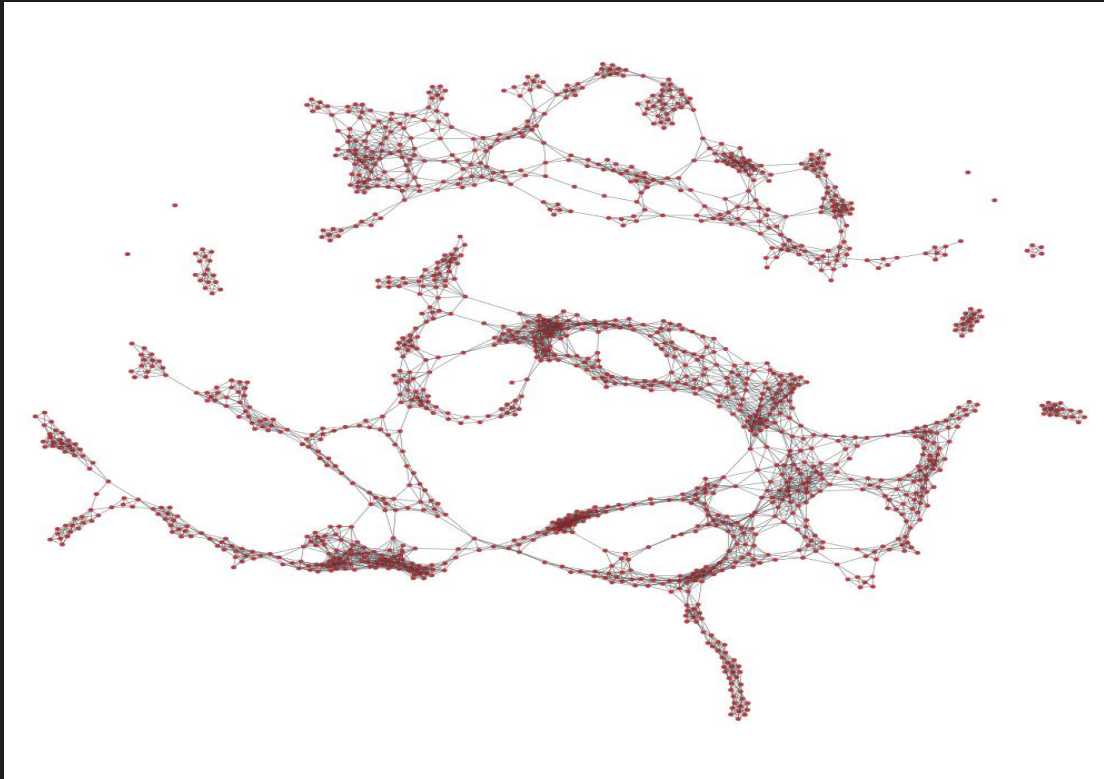
Some features:



Graph: Tumor 4:

- Ratio of the giant component :
$$R_{gc} = \frac{\text{Nodes in the biggest component}}{\text{Total number of nodes}} = 0.737$$
- Average closeness :
$$Cl_{avg} = \frac{1}{\sum_{i=1}^N \sum_{j=1, j \neq i}^N d_{ij}} = 441.73$$
- Average eccentricity (at 90%) :
$$Ec_{avg} = \frac{1}{N} \sum_{i=1}^N \max(d_{ij}, \forall j \in [1, N] - \{i\}) = 892.09$$

Some features:



Graph: Tumor 5:

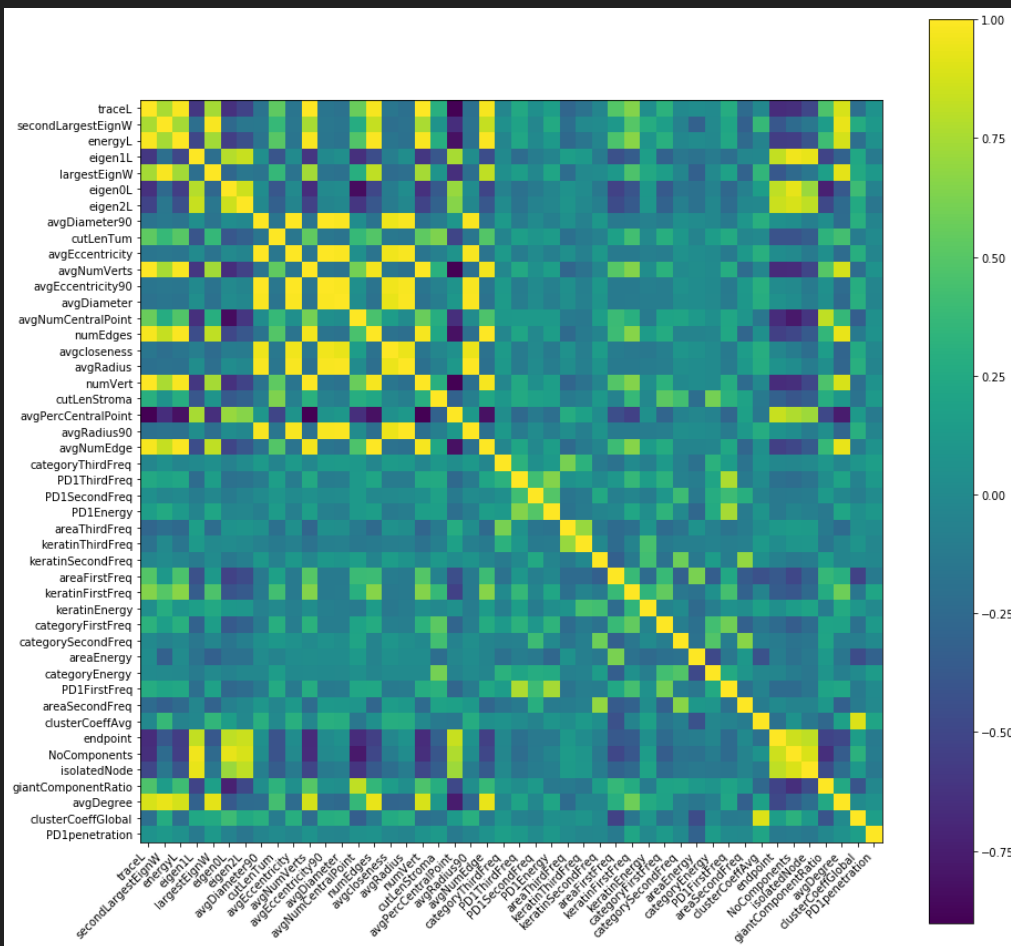
- Ratio of the giant component :
$$R_{gc} = \frac{\text{Nodes in the biggest component}}{\text{Total number of nodes}} = 0.675$$
- Average closeness :
$$Cl_{avg} = \frac{1}{\sum_{i=1}^N \sum_{j=1, j \neq i}^N d_{ij}} = 510.32$$
- Average eccentricity (at 90%) :
$$Ec_{avg} = \frac{1}{N} \sum_{i=1}^N \max(d_{ij}, \forall j \in [1, N] - \{i\}) = 1046.29$$

Correlation Matrices:

Correlation Matrix for the Full graph features of alive patients:

- Some correlations are trivia:
 - Distances features tend to have a strong correlation among themselves
- Some other need further analysis
 - We can see that frequencies representations of the signals seems to be more independent

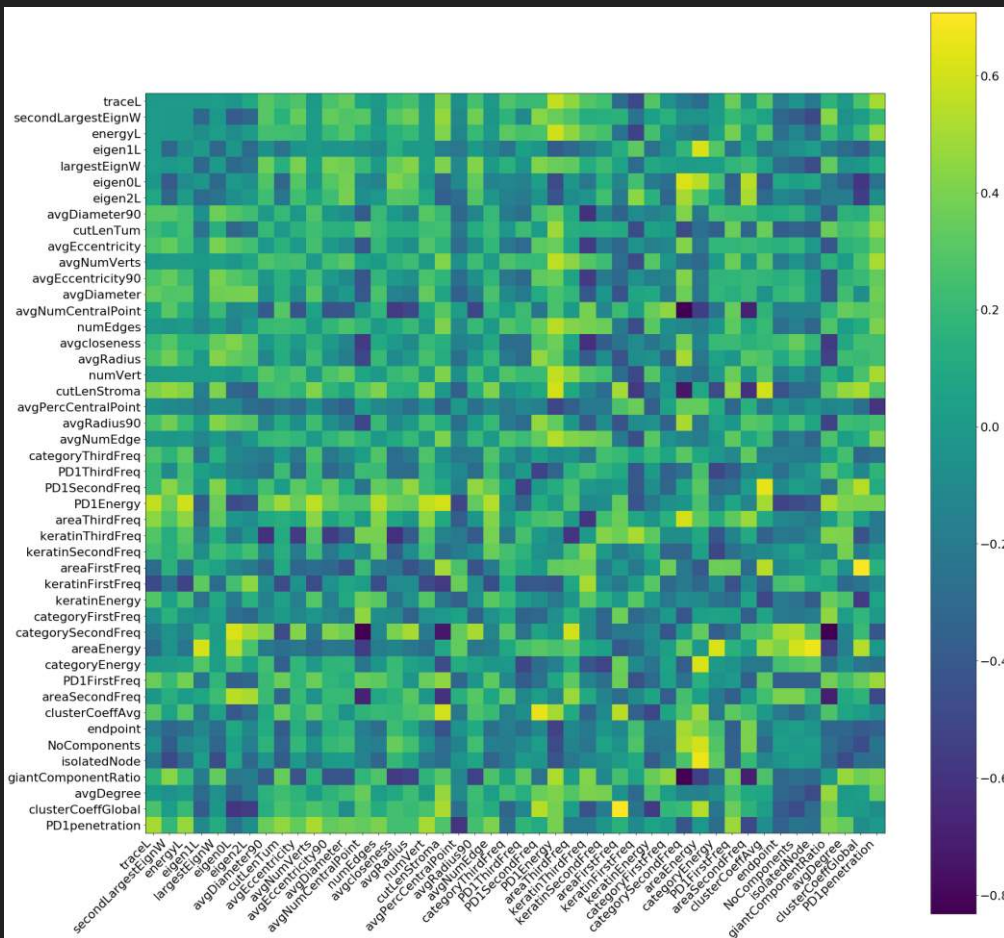
Correlation Matrices:



Correlation Matrix for the Full graph features of Dead patients:

- Some correlations are trivia:
 - Distances features tend to have a strong correlation among themselves
- Some other need further analysis
 - We can see that frequencies representations of the signals seems to be more independent

Correlation Matrices:



Correlation Matrix for the Full graph features, Difference:

- Need for a bigger dataset:
 - Even if we can see some high values in the difference of the correlation matrix, this may be caused by the small dataset
- Need for a more precise dataset:
 - The distinction “Dead or Alive” doesn’t give any hint on the state of the tumor, the time or a possible healing

Conclusion:

- Built graphs over cell images
- Computed 58 features over each graph
- Verified correlations between them
- Still need a validation with a machine learning technique
- Need for a bigger and more precise dataset
- More features and more analysis over them could be performed

References:

- Bülent Yener, " Cell-graphs: image-driven modeling of structure-function relationship", Rensselaer Polytechnic Institute, Troy, NY Communications of the ACM, Volume 60 Issue 1, January 2017
- Ulrike von Luxburg, "A Tutorial on Spectral Clustering", Max Planck Institute for Biological Cybernetics Spemannstr. 38, 72076 Tübingen, Germany, Statistics and Computing, 17 (4), 2007

Questions:

Some questions ?

Fin:

Thanks for your attention